Printed pages: 02 **Sub Code: EOE081/NOE081**

Paper Id **900153** **Roll No.** | | | | | | | | | | |

B. TECH.
(SEM-8) THEORY EXAMINATION 2018-19
NON CONVENTIONAL ENERGY RESOURSES

Time: 3 Hours ***Total Marks: 100***

Note: Attempt all Sections. If require any missing data; then choose suitably.

SECTION A

1. Attempt *all* questions in brief. 2 x 10 = 20

a. Discuss Energy conservation and Energy audit.
b. What is surface azimuth angle?
c. State seebeck Effect and peltier Effect.
d. Write the chemical reaction takes place in Alkaline Fuel Cell
e. What is an aerobic digestion?
f. Define Fill Factor.
g. Define fermentation in biomass energy?
h. What are the advantages of tidal power?
i. Are fossile fuels are renewable?
j. Define solar cell material?

SECTION B

2. Attempt any *three* of the following: 10 x 3 = 30

a. Describe solar photovoltaic (SPV) module with neat sketch?
b. What is fuel cell? Define working, operation and advantages?
c. Explain the working of horizontal axis two blade windmills with suitable diagram.
d. Explain with the help of the diagram, the principle of closed cycle ocean thermal energy conversion system?
e. Write short notes on: (i) Thermoelectric material (ii) Solar Cell Array

SECTION C

3. Attempt any *one* part of the following: 10 x 1 = 10

(a) Discuss the main features of various types of renewable and non-renewable energy sources. Also explain the importance of non-conventional energy sources in the context.
(b) Explain the mechanism of photoconduction in a PV cell.

4. Attempt any *one* part of the following: 10 x 1 = 10

(a) Define Solar Air Heater with neat sketch and also write its Application?
(b) Define Solar Air conditioning and Refrigeration System.

5. Attempt any *one* part of the following: 10 x 1 = 10

(a) Draw schematic diagram of an MHD power generating system having heat recovery steam generator. Explain the functioning of the steam.
(b) Describe the principle of working of H_2-O_2 Cell. Also give limitation.

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6. Attempt any *one* part of the following: 10 x 1 = 10

(a) What is the basic difference between thermoelectric and thermionic conversion systems? Also, explain the working of thermoelectric generators.

(b) What are the most favorable sites for installing wind turbines? Using Betz model of a wind turbine, derive the expression for power extracted from wind. Under what condition does the minimum theoretical power can be extracted from the wind turbine?

7. Attempt any *one* part of the following: 10 x 1 = 10

(a) Explain the 'Single Basin' and 'Two Basin' systems of tidal power. Discuss their advantages and limitations.

(b) Explain the process of gasification of solid biomass. What is the general composition of the gas produced and what is its heating value? What are its applications?

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